

# Eric Yu

Email: [eyu36@jhu.edu](mailto:eyu36@jhu.edu)

Website: <https://e7yu.github.io/>

## Education

- 2025–2029      **Johns Hopkins University**, Baltimore, MD, USA
- **B.S.**, Molecular & Cellular Biology and Chemistry
  - **GPA:** 4.00/4.00
  - Dean's List: Fall 2025, Spring 2026
- 2023–2025      **Georgia Institute of Technology**, Atlanta, GA, USA
- Dual Enrollment in High School
  - **GPA:** 4.00/4.00

My relevant coursework can be found at the end of my CV [here](#).

## Research Experience

- 2025–              **Research Assistant**, [Huang Lab](#), Johns Hopkins University
- *Advisor:* Prof. Xiongyi Huang
  - Engineering proteins via directed evolution to access new-to-nature biocatalytic reactions, including azidation, CF<sub>3</sub> azidation, and skeletal editing reactions, enabling pharmaceutically important transformations
  - Utilizing computational tools (PyMOL, AutoDock Vina) to analyze protein structure and substrate binding interactions
  - Preparing substrate libraries through organic synthesis to evaluate the substrate scope of evolved enzymes
- 2024–              **Research Assistant**, [Mao Lab](#), Indiana University School of Medicine
- *Advisor:* Prof. Weiming Mao
  - Designing a deep learning pipeline for automated segmentation and analysis of corneal endothelial cells and cross-linked actin networks with PyTorch
  - Assisting with literature review and the analysis of medical data, including those from optical coherence tomography (OCT), fundus imaging, and cell scratch assays

## Other Experience

- 2025–              **Violinist**, Hopkins Symphony Orchestra
- 2023–2025      **Violinist**, Atlanta Symphony Youth Orchestra

- Selected by a competitive multi-round audition process
  - Invited to perform at the 4<sup>th</sup> Annual Whitney E. Houston Foundation “Legacy of Love” Gala headlined by EGOT winner Jennifer Hudson
- 2022–2025      **Pianist & Violinist**, Georgia Philharmonic
- Performed Bernstein’s *Symphonic Dances from West Side Story* as a piano soloist
- 2021–2023      **Violinist**, Emory Youth Symphony Orchestra

## Awards & Honors

- 2025              **National Merit Scholar**, \$2,500 Scholarship
- Selected from a pool of over 15,000 National Merit Finalists
- 2021–2025      **4-time National Medalist**, Science Olympiad National Tournament
- 8x Top 10 national tournament placements
  - 4x State champion team and national qualifier
  - Top 6, 9, and 14 national team placements as a leading member and captain
- 2024              **High Honors**, U.S. National Chemistry Olympiad (USNCO)
- Top 50 out of over 16,000 participants nationally (top ~0.3%) in the most prestigious national chemistry competition
  - #1 in the state of Georgia on the Local Exam
- 2024              **Qualifier/Invitee**, American Invitational Mathematics Exam (AIME)
- Top scorer on the American Mathematics Competition (AMC)
- 2024              **Gold Award**, The President’s Volunteer Service Award
- Awarded to those with 250+ volunteer hours in a 12 month period
  - Previously won a silver award in 2022
- 2024              **Prize Winner**, Georgia Music Teacher’s Association
- Award of Excellence in both the Instrumental Concerto and String competitions
- 2021–2023      **First Prize**, Sonatina & Sonata International Youth Piano Competition
- Prize winner of the XIV and XV editions
  - 2x Carnegie Hall performance opportunities as winner

## Volunteering

- 2025–              **Event Supervisor**, Science Olympiad
- Test writer and grader for Maryland state and regional tournaments, Johns Hopkins invitational

- 2025–            **Musicare**
  - Music performance at local hospitals and nursing homes
- 2025–            **MedShare & SHARE**
  - Sorting medical supplies for distribution to developing nations
- 2025–            **Mural Art, Foundation for Hospital Art**
  - Painting murals to donate to hospitals and patients
- 2025–            **Circle K International**
  - Volunteering to help local communities in Baltimore
- 2022–2025      **Coach and Test Writer, Science Olympiad**
  - Coached my middle school team in physics events, guiding them to win the state championship and a national silver medal
  - Volunteered to write and grade tests for top invitational tournaments across the country

## **Skills**

### **Programming & Software**

- Programming Languages:            Python, MATLAB
- Machine Learning/AI:                PyTorch, scikit-learn
- Scientific & Technical Software:    PyMOL, ImageJ, AutoDock Vina, ChemDraw, Microsoft Office 365
- Typesetting:                            L<sup>A</sup>T<sub>E</sub>X

### **Laboratory-related**

- Organic Synthesis:                    Column chromatography, TLC, inert atmosphere (Schlenk line), liquid-liquid extractions, recrystallization, rotary evaporation
- Biomolecular Techniques:           Molecular cloning and library construction (PCR, mutagenesis, gel electrophoresis), bacterial culture & transformation (electroporation), protein expression & purification (sonication, affinity chromatography, SDS-PAGE), enzymatic assays
- Analytical Techniques:                Chromatography (GC-MS, HPLC), spectroscopy (NMR, IR, UV-Vis, CD)

## Publications

Detailed publication information can also be found at my [Google Scholar](#) .

## Journal Articles

- J1. Dai, J., Rayana, N. P., Peng, M., Sugali, C. K., Harvey, D. H., Dhamodaran, K., **Yu, E.**, Dalloul, J. M., Liu, S. & Mao, W. Inhibition of pterygium cell fibrosis by the Rho kinase inhibitor. *Sci. Rep.* **14**, 30930 (2024). [DOI](#).

## Peer-reviewed Conference Proceedings & Published Abstracts

- C1. Sugali, C. K., Dhamodaran, K., Dai, J., Harvey, D., Pushpala, N. M., **Yu, E.**, Huynh, K., Huang, A. Y. & Mao, W. Canonical Wnt signaling pathway inhibits glucocorticoid-induced ocular hypertension without compromising its anti-inflammatory function in the uveitis mouse model. *Invest. Ophthalmol. Vis. Sci.* **67**, 3010 (2026). [LINK](#).  
– 2026 [ARVO](#) Annual Meeting in Denver, CO, May 3–7, 2026.
- C2. Dai, J., Rayana, N. P., Peng, M., Sugali, C. K., Harvey, D. H., Dhamodaran, K., **Yu, E.**, Dalloul, J. M., Liu, S. & Mao, W. The Rho kinase inhibitor Y27632 inhibits unstimulated pterygium cell fibrosis. *Invest. Ophthalmol. Vis. Sci.* **66**, 6114 (2025). [LINK](#).  
– 2025 [ARVO](#) Annual Meeting in Salt Lake City, UT, May 4–8, 2025.

## Relevant Coursework

### Johns Hopkins University

- |                                              |                                              |
|----------------------------------------------|----------------------------------------------|
| – Physical Chemistry I (CHEM 301)            | – Genetics (BIOL 303)                        |
| – Physical Chemistry II (CHEM 302)           | – Biochemistry I (BPHY 315)                  |
| – Intmd. Organic Chemistry Lab (CHEM 228)    | – Protein Eng. & Biochemistry Lab (BPHY 253) |
| – Physical Chem. & Instrum. Lab (CHEM 307)   | – Biological Physics (PHYS 310)              |
| – Biological Models & Simulations (BMED 242) | – Contemporary Physics Seminar (PHYS 203)    |

### Georgia Institute of Technology *(dual enrolled in high school)*

- |                                        |                                               |
|----------------------------------------|-----------------------------------------------|
| – Cell & Molecular Biology (BIOS 3450) | – Linear Algebra (MATH 1554)                  |
| – Organic Chemistry I (CHEM 2311)      | – Multivariable Calculus (MATH 2551)          |
| – Organic Chemistry II (CHEM 2312)     | – Differential Equations (MATH 2552)          |
| – Synthesis Lab I (CHEM 2380)          | – Princ. & Appl. of Eng. Materials (MSE 2001) |
| – Intro to Modern Physics (PHYS 2213)  | – Intro to Computing (CS 1301)                |

---

Last updated: July 10, 2026